From Leisure to Loyalty: Behavioral Segmentation of Divvy Bike-Share Riders

A Divvy Bike-Share Conversion Analysis using R

Executive Summary

This analysis examines behavioral differences between casual riders and members of the Divvy bike-share system using trip data from Q1 2019 and Q1 2020. Results show that casual riders primarily use the service for longer, leisure-oriented trips concentrated on weekends, afternoons, and recreation-focused locations, while members exhibit consistent weekday commuting patterns centered around business districts and transit corridors.

Geospatial analysis further reveals a clear segmentation between recreational and commuter usage areas. Based on these insights, targeted strategies such as weekend-focused memberships, location-based marketing, and conversion incentives for repeat riders are recommended to increase casual-to-member conversion rates.

Business Task

Analyze the difference in bike usage patterns between casual riders and members to identify behavioral trends that marketing teams can use to design strategies that convert casual riders into members.

How do riding behaviors differ between casual riders and members across ride duration, timing, frequency, and station usage?

Data Sources

The data sources used in this analysis comes from Lyft Bikes and Scooters, LLC (“Bikeshare”) that operates the City of Chicago’s (“City”) Divvy bicycle service and has been made publicly available and licensed by Motivate International Inc.

Two quarterly datasets were used for this analysis: Divvy_Trips_2019_Q1 and Divvy_Trips_2020_Q1. These datasets contain data from January through March for each respective year. Each row in the dataset represents a single bike trip taken by a rider and includes information such as the start and end time of the trip, start and end station locations, and the usertype (casual/member).

Data Cleaning & Manipulation

Initial inspection revealed schema differences between datasets, requiring normalization before integration.

A check for duplicate records was done to ensure data integrity on both datasets and because there were columns on both sets that had columns with different names but with the same data, these were standardized to guarantee compatibility before integrating the datasets.

In the 2019 dataset, the columns “trip_id,” “from_station_id,” “from_station_name,” “to_station_id,” and “to_station_name” were standardized to “ride_id,” “start_station_id,” “start_station_name,” “end_station_id,” and “end_station_name” respectively.

In the 2020 dataset, the columns “started_at,” “ended_at,” and “member_casual” were standardized to “start_time,” “end_time,” and “usertype” respectively.

In 2019, the values of the “usertype” column “Customer” and “Subscriber” were transitioned to “casual” and “member” respectively to match the values of the 2020 dataset “usertype” column and to improve data quality.

The columns “rideable_type,” “tripduration,” “gender,” and “birthyear” were removed because this data was not in the 2020 dataset and the column “bikeid” was removed from the 2020 dataset since there is no matching column on the 2019 dataset.

Rows with blank start/end stations were eliminated.

New variables “ride_length” and “day_of_week” were created on both datasets to support behavioral analysis of riders.

“ride_length” column was calculated to measure the duration of each trip.

“day_of_week” was created with a number for the day of the week (1-Sunday/7-Saturday), to allow analysis of riding patterns by that day of the week.

Lastly, the columns to hold latitude and longitude data were created on the 2019 dataset to migrate this data from the 2020 dataset to map route trends for visual comparison of casual vs. member routes.

After finishing these preparation steps, the datasets were loaded to RStudio, mutated the datasets because some of the cleaning changes were reverted or were needed, like adding the latitude and longitude columns to be combined into a single dataset for further analysis.

In RStudio, all the required packages were installed to start the analysis of the data. The packages used are tidyverse (Meta-package), janitor (cleaning), plotly (interactive plots), leaflet (mapping), scales (map scaling), and lubridate (time analysis).

Analysis Process

Ride Length Analysis:

- Calculated the average, and median of all the rides in the dataset and pulled a summary of the data for an overview of the data I'm working on.
- Created a summary of the data by ride length and calculated the ride length quantiles.
- Found the distribution of the ride length to see where most rides fall and where the extreme rides begin to determine where to filter and further clean the dataset.
- After finding where the most rides fall, I decided to focus on a span of 24 hours to make it more concise and because outside this timeframe, they may represent false starts, docking errors, or irregular records that do not represent normal rider behavior.
- Plotted the finding of the summary on ride length.

Day of the Week Analysis

- Mutated the day_of_week column to reflect the name of the day and make it more readable.
- Created a summary of rides by day to see which days were the most busy for both usertype.
- Found and normalized the percentages of rides by day by rider type.
- Plotted the results.

Hour of the Day Analysis

- Created the column hour to find which hours of the day were the most busy by both rider types.
- Calculated average rides per hour.
- Plotted the results grouping them by usertype and fixed the labels to reflect am/pm format.

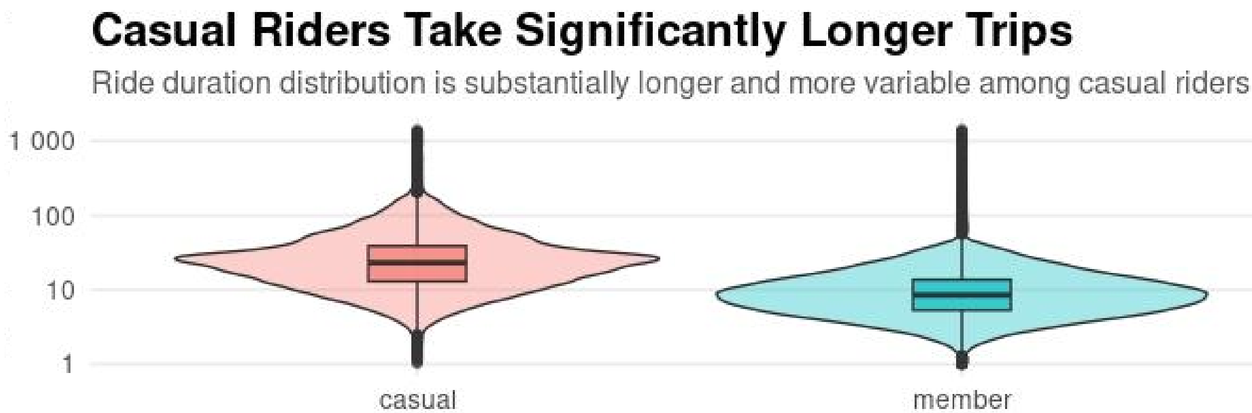
Station Analysis

- Found the top start stations as well as the top end stations for both rider types.
- Calculated the percentages of round trips .
- Created visualizations to further support my hypothesis.

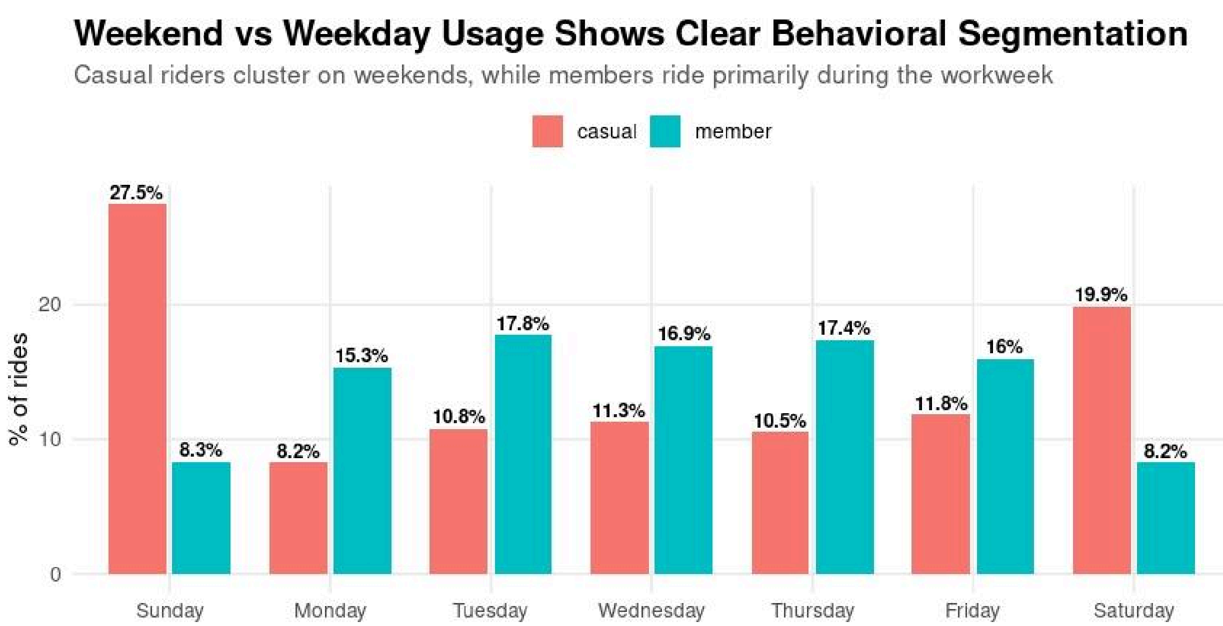
Key Findings

- **What is the difference in ride length between casual riders and members?**

Casual riders tend to take longer and less consistent rides, while members take shorter, more routine trips.

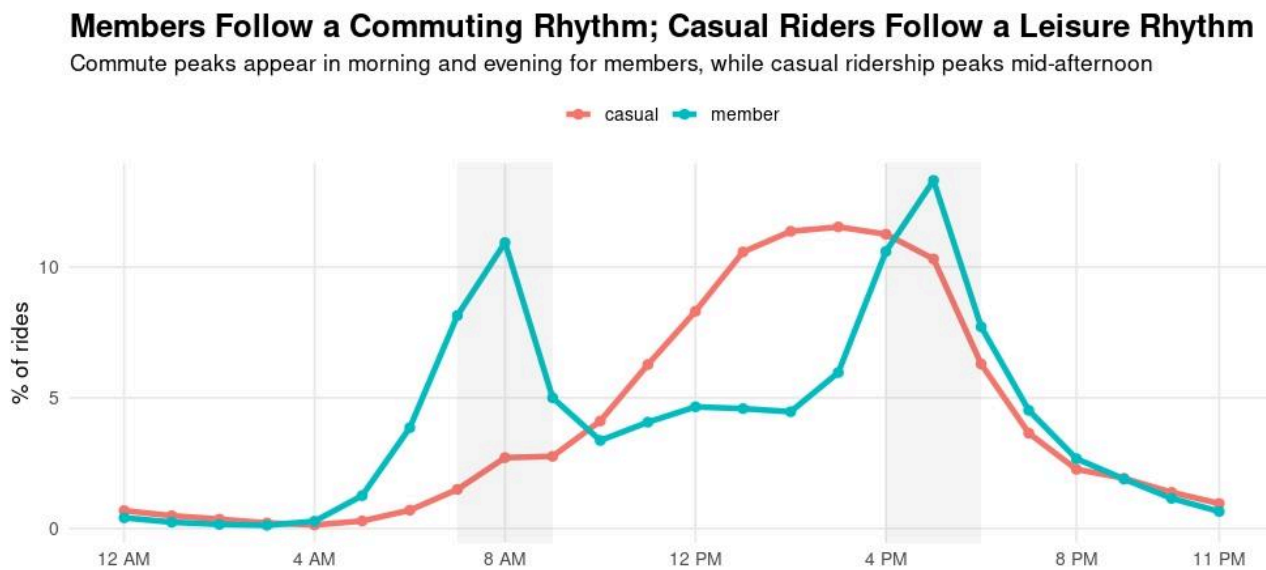


- **How does riding behavior across the week differ between casual riders and members?**
  While the strong weekend usage pattern of casual riders shows that nearly half of the rides occur on Saturdays and Sundays, suggesting primarily recreational usage; the members ride most frequently during weekdays, especially Tuesday through Thursday, indicating routine commuting behavior.



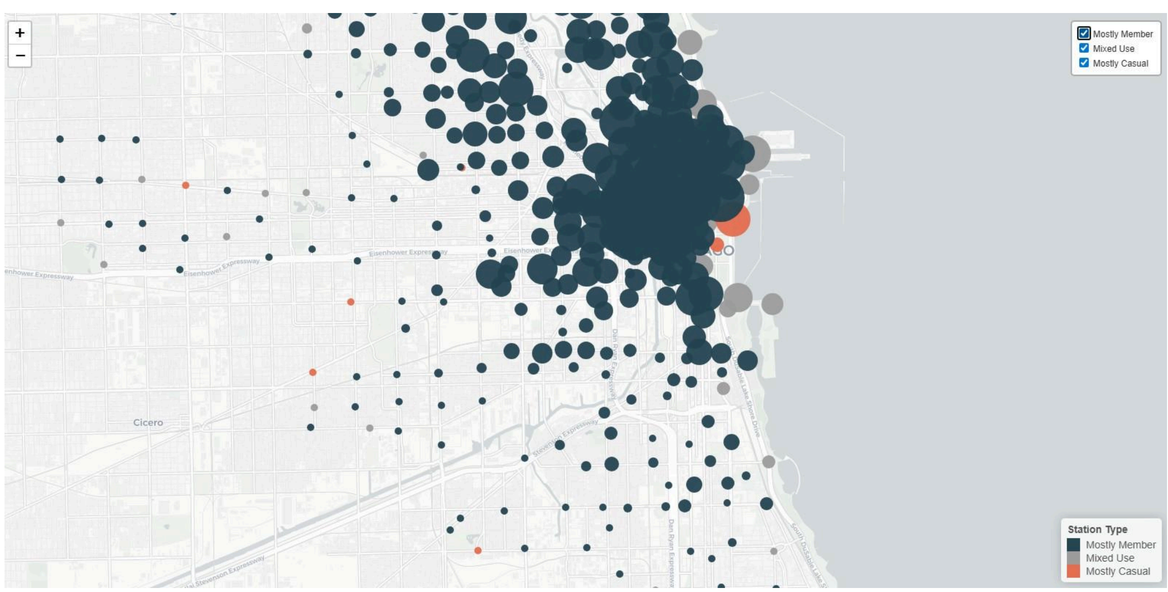
- **At what times of the day do members vs casual riders ride the most?**
  Members with strong ride spikes during morning commuting hours and again during evening commuting hours suggest that members primarily use Divvy as a transportation solution for work related commuting. Casual riders, by contrast, show gradual increase in trips in the late

morning peaking during the afternoon between 2 and 4 pm, and declining in the evening suggesting recreational, social, and leisure-oriented riding behavior.

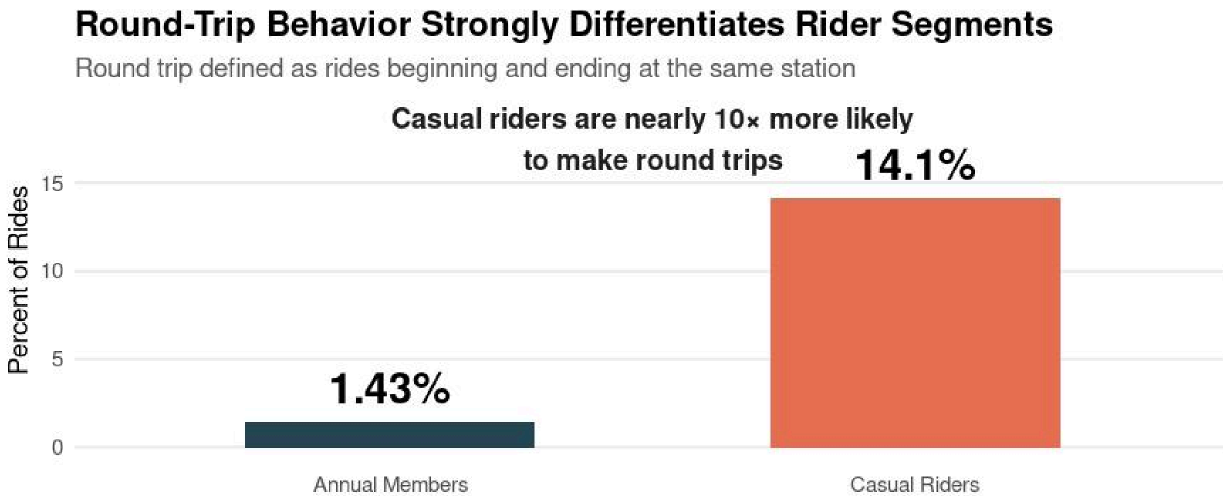


- **Which stations are most popular for casual riders vs members?**

Casual riders cluster around Chicago's lakefront, parks, attraction-adjacent stations, and recreation-focused corridors, while members concentrate around downtown business districts, office centers, and transit hubs. Geospatial mapping highlights a clear separation between leisure-oriented and commuter-oriented usage zones.



- **How does round-trip behavior differ between rider types?**
  Casual riders are nearly 10x more likely to make round trips (14.1% vs 1.43%), suggesting greater use for recreational loops, sightseeing, and leisure outings, while members overwhelmingly display point-to-point transportation behavior.



Recommendations

1. Pilot a Weekend-Focused Membership or Ride Bundle

Test a lower-cost weekend-focused plan or ride-credit bundle targeted specifically toward repeat casual riders. This strategy aligns with observed recreational riding behavior while minimizing the risk of existing members downgrading from full memberships.

2. Target High-Concentration Leisure Stations for Conversion Campaigns

Deploy in-app and on-location marketing at high-traffic recreational stations such as lakefront and attraction-adjacent areas. Focus messaging on frequent weekend riders with prompts such as:
"You've ridden multiple weekends this month — save with a weekend/annual membership."

3. Develop Personalized Engagement Campaigns for High-Frequency Casual Riders

Use ride-frequency and behavioral data to identify casual riders who exhibit repeated weekend or recreational usage patterns. Deliver targeted in-app notifications, email campaigns, or app-based usage summaries that highlight potential cost savings and membership benefits based on their riding habits.

Limitations

- The data available was only Q1 data which limited a seasonal analysis.
- There were no enough demographic data available to do demographic targeted analysis.
- Due to the lack of rider's id as a unique identifier, repeat casual riders could not be identified.
- Station context is inferred since only the dataset from 2020 has the stations' coordinates.

Next Steps

- Request yearly data for full-year seasonality analysis.
- Implement A/B testing campaigns.